EE4-51
EE9-FPN2-11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

EEE PART IV: MEng and ACGI

POWER SYSTEM ECONOMICS

Tuesday, 27 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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The Questions

Question 1

(a) Describe the significance of the concept of system marginal price and explain why it varies with time in the case of electricity.

[2]

(b) Three generators are available to supply demand of $D = 600\ [MW]$. The cost of generating power $C_i(P_i)$ corresponding to each of the generators is:

$$C_1(P_1) = 800 + 2 * P_1 + 0.05 * P_1^2\ [\frac{£}{h}]$$

$$C_2(P_2) = 650 + 8 * P_2 + 0.25 * P_2^2\ [\frac{£}{h}]$$

$$C_3(P_3) = 500 + 16 * P_3 + 0.55 * P_3^2\ [\frac{£}{h}]$$

Calculate the optimal production of each generator, the system marginal cost, the system average cost and the profits of each generator.

[6]

(c) How would the dispatch change if three generators have the opportunity to buy some of this energy on the spot market at a price of 42.60 £/MWh? How much energy will be bought from the spot market?

[3]

(d) Assume that the minimum and maximum output limit of each generator is:

$$P_1^{min} = 100\ [MW],\ P_1^{max} = 600\ [MW]$$

$$P_2^{min} = 60\ [MW],\ P_2^{max} = 120\ [MW]$$

$$P_3^{min} = 40\ [MW],\ P_3^{max} = 100\ [MW]$$

(i) Determine the optimal production of each generator, the system marginal cost, the system average cost and the profits of each generator.

[4]

(ii) Based on the results, briefly explain the impact of considering these maximum output limits on the total cost of operating the system and the profits of the three generators.

[2]

(e) In the solution of **(d)**, the profit of one of the generators is negative. Explain why this happens, what this means for this generator, as well as why and how it should be avoided in practice.

[3]

Question 2

(a) In the context of the competitive market concept, answer the following questions:

(i) What are the conditions for perfect competition?

[2]

(ii) How can a company in an imperfect competition environment influence the market price?

[1.5]

(iii) What is the relationship between the marginal cost of production and the market price in perfectly and imperfectly competitive markets?

[1.5]

(b) Consider a market for electrical energy that is supplied by two generating companies with the two corresponding cost functions:

$$C_A(P_A) = 26 * P_A + 0.5 * P_A^2 \ [\frac{£}{h}]$$

$$C_B(P_B) = 30 * P_B + 0.3 * P_B^2 \ [\frac{£}{h}]$$

The inverse demand curve for this market is estimated to be:

$$\pi = 200 - 1.5 * D \ [\frac{£}{MWh}]$$

Assuming perfect competition, calculate:

(i) electricity price and level of demand

[1.5]

(iii) production levels and the profits made by the generating companies.

[1.5]

(c) Assuming a Cournot model of competition:

(i) Form a table to calculate the Nash equilibrium point of this market, i.e. market price, demand quantity and profit of each company for different levels of productions for each of the companies. Consider range of productions of company A at 35MW, 37MW and 39MW and of company B at 37MW, 39MW and 41MW.

[8]

(ii) Based on the results, briefly explain the impacts of imperfect competition on the market outcome.

[2]

(d) Explain the principal difference between Bertrand and Cournot competition models.

[2]

Question 3

For the system shown in Figure 1, and the accompanied data given in Tables 1, 2 and 3, answer the following questions:

(a) Ignoring the impact of transmission constraints, calculate：

(i) minimum cost dispatch of the generators

[2]

(ii) the corresponding network flows and show that line 1-2 would be overloaded.

[2]

(b) This line overload can be eliminated by *A)* increasing generation at node 3 and reducing generation at node 1; or by *B)* increasing generation at node 2 and reducing generation at node 1.

(i) Calculate the hourly cost of options *A)* and *B)* above. Which of the two options is preferable and why?

[6]

(ii) Calculate the nodal prices when the more preferable of the two options compared in **(i)** is implemented.

[4]

(c) In order to manage the exposure to price fluctuations, Consumer (Load) connected to bus 2 has made a contract for difference with generation at bus 1 at a strike price of 26.5£/MWh and purchased a transmission congestion contract that gives the right to transport 90MW from bus 1 to bus 2. Calculate:

(i) income from the transmission congestion contract received by the Consumer (Load) at bus 2

[3]

(ii) net cost and corresponding unit price that demand at bus 2 will pay for the energy consumed under this condition.

[3]

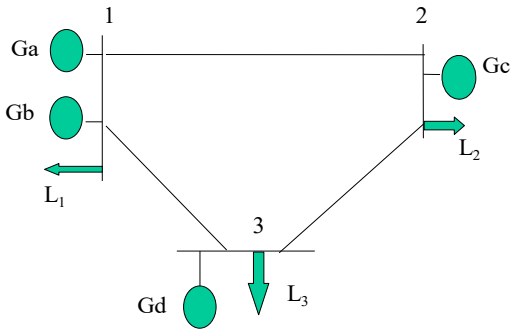


Figure 1: System structure

Table 1 Load data

Node	Load (MW)
1	80
2	90
3	330

Table 2 Generator data

Generator	Capacity (MW)	Marginal cost (£/MWh)
Ga	170	19
Gb	345	15
Gc	110	35
Gd	105	25

Table 3 Line data

Line	Per unit reactance	Capacity (MW)
1-2	0.3	160
1-3	0.3	260
2-3	0.2	110

Question 4

(a) Two areas, A and B, of the Borduria power system are connected by a 1200MW transmission link operated by the Borduria Transco. System load is concentrated in area B. In winter, the load is 3,500MW while summer load is 2,000 MW. The cost of generation in the areas can be modelled by the following expressions:

$C(P_A) = 1 \cdot P_A + 0.001 \cdot P_A^2$ [£/h] for area A

$C(P_B) = 2 \cdot P_B + 0.002 \cdot P_B^2$ [£/h] for area B

Determine the optimal levels of generation in areas A and B for each of the seasons neglecting the impact of the transmission system.

[4]

(b) If necessary, modify the levels of generation computed in (a) to take into consideration the capacity of the transmission link. For each of seasons:

(i) What are the electricity prices in areas A and B?

[2]

(ii) Calculate the generation costs in areas A and B.

[2]

(c) Assuming that the duration of the winter and summer period is 2,700h and 6,060h respectively, calculate:

(i) The total annual generation costs

[2]

(ii) The total annual revenue received by the generators and the charge imposed on demand if electricity is priced at locational short-run marginal prices.

[2]

(iii) The total annual transmission revenue received by Borduria Transco in this case.

[2]

(d) Borduria Transco has proposed to the Borduria Regulator to reinforce the line between the two areas by 900 MW. Assuming that the annuitised investment cost of the additional line is 5,000,000 £/year, determine if this proposed investment is justified.

[6]